IDS 702 Project Proposal

Agastya Vinchhi, Steven Xu, Zhuotong Xie, Gulnar Gasimzade

Project Proposal/Statistical Analysis Plan

Link to your team's Github Repo: https://github.com/agastyavinchhi/Statistics-Final-Project

Dataset Option 1

Project Title: Confidence in Science: Evidence from the 2024 General Social Survey

Data citation (use APA formatting or something similar): Davern, M., Bautista, R., Freese, J., Herd, P., & Morgan, S. L. (2025). *General Social Survey 2024* [Data set]. NORC at the University of Chicago. Sponsored by the National Science Foundation. https://gss.norc.org

Original data collection date (when was the dataset published?): Data were collected in 2024 and released by NORC in 2025

Data pull date (when did you download the data you will use?): November 4, 2026

---

1 Research Question

Write the research question you are interested in investigating in your analysis. Consult this helpful guide for writing a good research question. Go beyond "is there a relationship between X and Y" to incorporate some statistical complexity, e.g., "is there a relationship between X and Y and does this relationship change over time?" You may also want to provide a second RQ as a backup and I can provide feedback on both options. Remember that you are required to use an inferential research question.

Among adults surveyed in the 2024 General Social Survey (GSC), is more education associated with greater confidence in the scientific community, and does the strength of this association differ by political party identification (interaction term), after adjusting for age, sex, race, region, family income, and religious service attendance?

2 Study Population

Describe the observation units and how the data were collected.

The observation units are individual adults who are 18+ and are from the United States. They are adults who responded to the 2024 GSS survey. This data was collected through address-based probability sampling. Most respondents completed this survey online, and the rest was in person or by phone. The question on if the person is confident in the scientific community was asked at random to two-thirds of the individuals. From 3986 respondents, that reduces the data to 2565 valid responses to answer our research question.

3 Outcomes, Exposures, and Additional Variables of Interest

Add rows as necessary. Use the data dictionary for the coding specifications, e.g., 1=Diagnosed with CHD, 0=No diagnosis

3.1 Primary Outcome(s)

Variable Name	Description	Coding specification
consci	Confidence in the scientific community	Ordinal: 1 = a great deal, 2 = only some, 3 = hardly any

3.2 Primary Independent Variable(s)

Variable Name	Description	Coding specification
degree	Highest degree completed	Ordinal: 0 = less than high school, 1 = high school, 2 = associate/junior college, 3 = bachelor's, 4 = graduate
partyid	Political party identification	Categorical: Democrat (0–2), Independent (3), Republican (4–6) other party

3.3 Additional Variables of Interest (including potential confounders)

Variable Name	Description	Coding specification
age	Age of respondent	Continuous variable
realinc	Family income (constant dollars)	Continuous
childs	Number of children	Continuous (count), 0–8 (8 = eight or more)
maeduc	Mother's highest year of school completed	Continuous, 0–20 years
tvhours	Hours per day watching TV	Continuous, 0–24 hours
sex	Sex of respondent	Binary: 1 = male, 2 = female
race	Race of respondent	Categorical: 1 = white, 2 = Black, 3 = other
region	Region of interview	Categorical: 1 = Northeast, 2 = Midwest, 3 = South, 4 = West
attend	Religious service attendance	Ordinal: 0 = never through 8 = several times a week
polviews	Political ideology (liberal to conservative)	Ordinal: 1 = extremely liberal through 7 = extremely conservative
wrkstat	Labor force status	Categorical: 1 = full time, 2 = part time, 3 = temporarily not working, 4 = unemployed, 5 = retired, 6 = in school, 7 = keeping house, 8 = other

4 Statistical Analysis Plan

4.1 Exploratory Data Analysis

Detail how you will investigate the data, including basic characteristics of the outcome and independent variable(s) of interest (e.g., "The outcome is continuous, so we will generate a histogram of the outcome. Then, we will generate a stacked histogram/scatter plot to explore the relationship between the outcome and the independent variable")

To evaluate our data, we will tailor our exploratory visualizations to the categorical structure of our variables. Our outcome, `consci` (confidence in science), is an ordinal categorical variable with three ranked tiers: "Hardly Any", "Only Some", and "A Great Deal". Because the outcome consists of discrete ordered categories we will use a bar chart to inspect the overall distribution of responses and identify any class imbalances.

For our important predictors for analysis, `degree` (education) and `partyid` (political affiliation), we will construct a stacked proportional bar chart to examine confidence levels across educational

tiers. To explore our interaction term better (Degree × Party ID), we will facet these bar charts across political affiliations, allowing us to visually compare whether scientific trust differs between Democrats and Republicans, organized by education.

4.2 Statistical Model (write in mathematical notation and define each term)

$$\begin{aligned}\text{logit}[P(\text{Confidence in Science} \leq j)] = \beta_{0j} - (&\beta_1 \,\text{Highest Degree} + \beta_2 \,\text{Political Party} \\ &+ \beta_3 \,(\text{Highest Degree} \times \text{Political Party}) \\ &+ \beta_4 \,\text{Age} + \beta_5 \,\text{Family Income} \\ &+ \beta_6 \,\text{Number of Children} + \beta_7 \,\text{Mother's Years of Education} \\ &+ \beta_8 \,\text{Daily TV Hours} + \beta_9 \,\text{Sex} + \beta_{10} \,\text{Race} \\ &+ \beta_{11} \,\text{Region} + \beta_{12} \,\text{Religious Attendance} \\ &+ \beta_{13} \,\text{Political Views} + \beta_{14} \,\text{Work Status})\end{aligned}$$

4.2 Rationale for the statistical model (why are you choosing to use this type of model? What about your data structure/characteristics are you accommodating with this model type?)

We chose an ordinal logistic regression model because our outcome, `consci` (confidence in science), is an ordinal categorical variable with three ranked levels: "Hardly Any", "Only Some", and "A Great Deal".

Linear regression would not work well here because it assumes our outcome is continuous, while multinomial regression would ignore the order of the outcome.

4.3 Model Assumptions

For our ordinal logistic regression model, we will rely on a few key assumptions:

Proportional odds (parallel slopes): We assume that the effect of each predictor (such as education and party affiliation) is the same across each step of the confidence levels.

Independence of observations: Each respondent's answer is assumed to be independent of others, which aligns with how the GSS survey was designed and collected.

Linearity for continuous predictors: For numeric variables like age and family income, we assume a linear relationship with the log-odds of the outcome.

No multicollinearity: We assume our predictors and demographic control variables are not overly correlated with each other.

4.4 Model Assessment (how will you assess the assumptions and model performance?)

To check these assumptions and evaluate how well our model fits the data in our final report, we will take the following steps:

Evaluating the proportional odds assumption: We will visually check the empirical log-odds across separate cutpoints to see if the slopes look roughly parallel, or compare our model fit with a less restrictive model.

Checking multicollinearity: We will examine Variance Inflation Factors (VIF) for all variables, ensuring that adding the interaction term (Degree $\times$ Party ID) doesn't introduce severe collinearity (keeping VIF values below 5).

Checking continuous variables: We will plot residuals and look for non-linear patterns in age and income to make sure their linear form is appropriate.

Assessing overall model performance: We will assess model fit using the Likelihood Ratio Test, compare nested model versions using AIC / BIC, and check predicted probabilities to see how accurately the model captures people's trust in science.

Exploratory Data Analysis

First, provide a snapshot of the data using the glimpse function or something similar. Then, present your EDA according to your SAP above. At minimum, you should present basic characteristics of your outcome alone (descriptive statistics and a plot) and at least one key relationship of interest, but the EDA should not exceed 3 pages. It may be useful to present a "Table 1" that gives descriptive statistics for several variables of interest. You are not required to complete data cleaning for the EDA.

```
# A tibble: 6 x 14
  consci degree partyid sex   race  region attend polviews wrkstat   age realinc
  <fct>  <fct>  <fct>   <fct> <fct> <fct>  <fct>  <fct>    <fct>   <dbl>   <dbl>
1 hardl~ bache~ not ve~ male  black north~ every~ moderat~ workin~    33  43560
2 a gre~ gradu~ strong~ male  white north~ never  slightl~ retired    64  24502.
3 a gre~ assoc~ strong~ fema~ white north~ never  extreme~ retired    69  43560
4 iap    high ~ indepe~ male  other north~ sever~ don't k~ workin~    19  58080
5 only ~ high ~ indepe~ fema~ white north~ nearl~ moderat~ retired    70  24502.
6 a gre~ assoc~ indepe~ male  other north~ 2-3 t~ skipped~ unempl~    53   2722.
# i 3 more variables: childs <dbl>, maeduc <dbl>, tvhours <dbl>
```

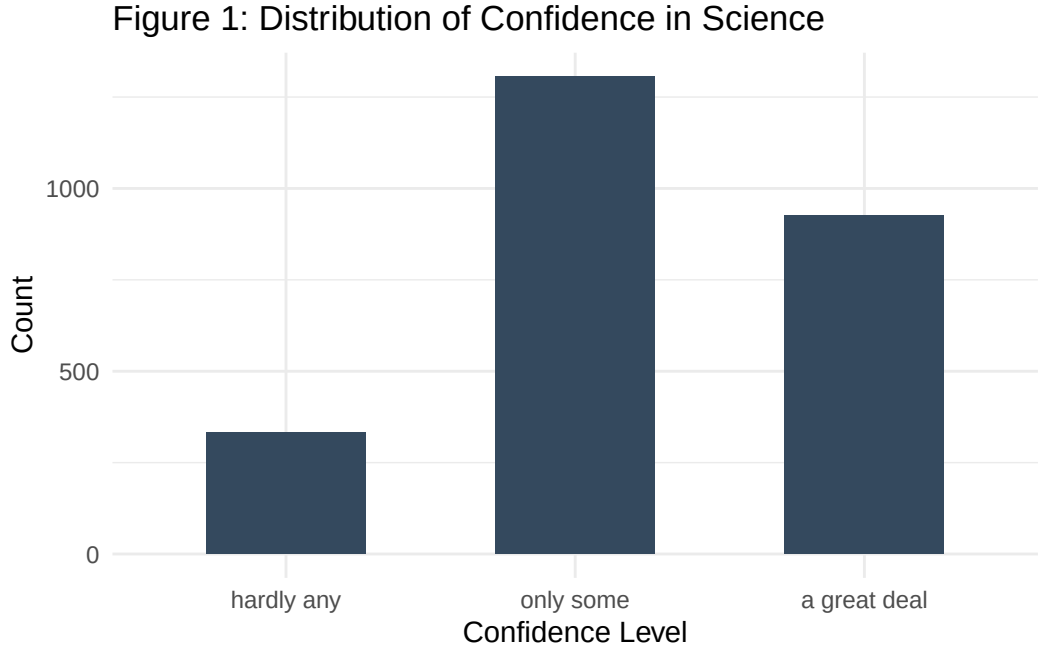
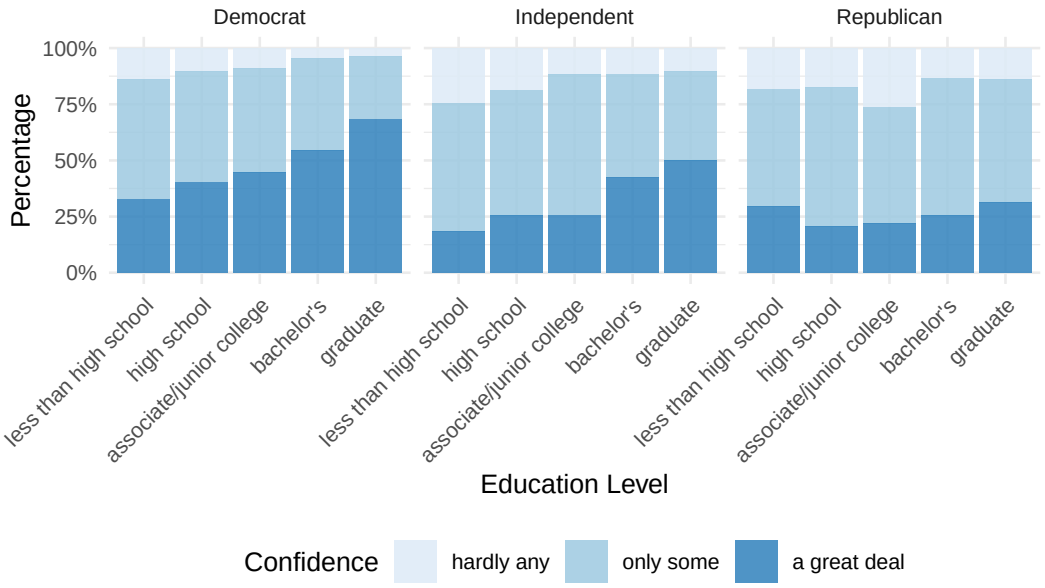


Figure 2: Confidence in Science across Education by Political Party



Dataset Option 2

Project Title:

The Association Between Weather Conditions and Forest Fire Burned Area: The Moderating Role of Relative Humidity

Data citation (use APA formatting or something similar):

Cortez, P., & Morais, A. (2007). *Forest Fires* [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5D88D

Original data collection date (when was the dataset published?):

2000–2003 (collected); 2008 (published in the UCI Repository)

Data pull date (when did you download the data you will use?):

2026 /11/06

---

1 Research Question

Write the research question you are interested in investigating in your analysis. Consult this helpful guide for writing a good research question. Go beyond "is there a relationship between X and Y" to incorporate some statistical complexity, e.g., "is there a relationship between X and Y and does this relationship change over time?" You may also want to provide a second RQ as a backup and I can provide feedback on both options. Remember that you are required to use an inferential research question.

Does relative humidity moderate the association between temperature and forest fire burned area, after adjusting for wind speed and seasonal variation?

2 Study Population

Describe the observation units and how the data were collected.

The dataset contains 517 forest fire records collected from Montesinho Natural Park in north-eastern Portugal between 2000 and 2003. Each observation represents a recorded forest fire and includes information on burned area, meteorological conditions, fire weather indices, and spatial and temporal characteristics.

The data were obtained from the UCI Machine Learning Repository and were originally compiled by Cortez and Morais (2007). As this is an observational dataset, the analysis will focus on statistical associations rather than causal relationships. The findings may not generalize to forest fires in other geographical regions or time periods.

3 Outcomes, Exposures, and Additional Variables of Interest

3.1 Primary Outcome(s)

Variable Name	Description	Coding specification
area	Forest area burned by each recorded fire	Continuous, measured in hectares (ha), ranging from 0 to 1090.84
log_area	Log-transformed burned area, used as the regression response	Continuous, calculated as log(area + 1)

3.2 Primary Independent Variable(s)

Variable Name	Description	Coding specification
temp	Ambient temperature	Continuous, measured in degrees Celsius (°C)
RH	Relative humidity	Continuous, measured as a percentage (%)

An interaction term between temperature and relative humidity will be included to investigate whether the association between temperature and burned area varies across different humidity levels.

3.3 Additional Variables of Interest (including potential confounders)

Variable Name	Description	Coding specification
wind	Wind speed	Continuous, measured in km/h
month	Month in which the fire was recorded	Categorical, Jan–Dec
season	Season in which the fire was recorded	Categorical, derived from month: Winter (Dec–Feb), Spring (Mar–May), Summer (Jun–Aug), Fall (Sep–Nov)

4 Statistical Analysis Plan

4.1 Exploratory Data Analysis

We will begin by examining the structure of the dataset and calculating descriptive statistics, including the mean, median, standard deviation, and range for burned area, temperature, relative humidity, and wind speed. We will also examine the number of observations across different seasons.

Next, we will visualize the distribution of burned area using histograms to identify skewness, zero values, and potential outliers. Since burned area is highly right-skewed, we will also examine the distribution after applying a log(area + 1) transformation.

To explore the potential interaction between temperature and relative humidity, we will create a scatter plot of temperature against log-transformed burned area, with observations grouped into higher and lower humidity levels based on the median relative humidity. We will add separate fitted regression lines for each group to visually examine whether the relationship between temperature and burned area differs across humidity levels.

4.2 Statistical Model (write in mathematical notation and define each term)

We plan to use multiple linear regression with an interaction term to examine whether relative humidity moderates the association between temperature and forest fire burned area. Because burned area is highly right-skewed, the response variable will be transformed using log(area + 1).

Temperature and relative humidity will be mean-centered to improve the interpretation of the regression coefficients. Wind speed and season will be included as additional covariates, with winter serving as the reference category.

The proposed model is:

$$\begin{aligned}\log(1 + \text{area}_i) = \beta_0 &+ \beta_1 T_i + \beta_2 H_i \\ &+ \beta_3 (T_i H_i) + \beta_4 W_i \\ &+ \gamma_1 S_{1i} + \gamma_2 S_{2i} \\ &+ \gamma_3 S_{3i} + \epsilon_i\end{aligned}$$

where:

- T_i is the mean-centered temperature.
- H_i is the mean-centered relative humidity.
- $T_i H_i$ is the interaction between temperature and relative humidity.
- W_i represents wind speed.
- S_{1i}, S_{2i}, and S_{3i} are indicators for spring, summer, and fall, respectively, with winter as the reference category.
- ϵ_i is the error term.

The primary parameter of interest is β_3, which measures how the association between temperature and log-transformed burned area changes with relative humidity.

4.2 Rationale for the statistical model (why are you choosing to use this type of model? What about your data structure/characteristics are you accommodating with this model type?)

Multiple linear regression is appropriate for this analysis because our response variable, log-transformed burned area, is continuous, and we aim to examine its association with several explanatory variables simultaneously.

We will include an interaction term between temperature and relative humidity to investigate whether the association between temperature and burned area varies across different humidity levels. Wind speed and season will be included as additional covariates to account for differences in weather conditions and seasonal patterns.

Since the original burned area variable is highly right-skewed and contains extreme values, we will apply a log(area + 1) transformation to reduce the influence of extreme observations. However, the substantial number of zero values may still present challenges for linear regression.

We will therefore evaluate the model assumptions and assess whether the proposed model is appropriate for the data.

4.3 Model Assumptions

Our multiple linear regression model relies on the following assumptions:

1. **Linearity:** The expected log-transformed burned area is a linear function of the specified predictors and their interaction term.
2. **Independence:** Observations and their error terms are assumed to be independent.
3. **Homoscedasticity:** The variance of the error terms is assumed to be constant across different predictor values.
4. **Normality of Errors:** The error terms are assumed to be approximately normally distributed for conventional statistical inference.
5. **No Perfect Multicollinearity:** The predictors should not exhibit perfect linear relationships with one another.

We also assume that the proposed model is reasonably specified and that extreme observations do not unduly determine the estimated relationships.

4.4 Model Assessment (how will you assess the assumptions and model performance?)

We will evaluate the regression model using diagnostic plots and statistical measures. Residuals versus fitted values plots will be used to assess the linearity assumption and identify potential heteroscedasticity, while Q-Q plots will be used to examine the normality of residuals. Variance inflation factors (VIF) will be calculated to assess multicollinearity among predictors.

We will also examine leverage and Cook's distance to identify potentially influential observations, particularly those associated with extremely large burned areas. We will consider possible temporal and spatial dependence among observations, although the available data may limit a formal assessment of independence.

Model fit will be evaluated using R-squared and adjusted R-squared. Our primary statistical inference will focus on the temperature–humidity interaction coefficient, using its estimated value, 95% confidence interval, and p-value to assess whether the association between temperature and burned area varies across humidity levels.

If substantial heteroscedasticity or influential observations are detected, we will consider robust standard errors and sensitivity analyses to evaluate the stability of our findings.

Exploratory Data Analysis

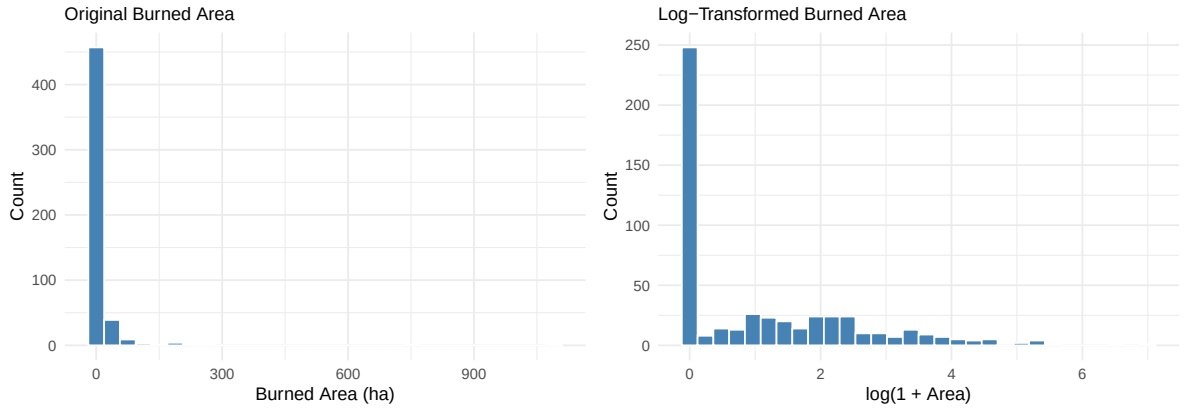
```
Rows: 517
Columns: 13
$ X     <dbl> 7, 7, 7, 8, 8, 8, 8, 8, 8, 7, 7, 7, 6, 6, 6, 6, 5, 8, 6, 6, 6, 5~
$ Y     <dbl> 5, 4, 4, 6, 6, 6, 6, 6, 6, 5, 5, 5, 5, 5, 5, 5, 5, 5, 4, 4, 4, 4~
$ month <chr> "mar", "oct", "oct", "mar", "mar", "aug", "aug", "aug", "sep", "~
$ day   <chr> "fri", "tue", "sat", "fri", "sun", "sun", "mon", "mon", "tue", "~
$ FFMC  <dbl> 86.2, 90.6, 90.6, 91.7, 89.3, 92.3, 92.3, 91.5, 91.0, 92.5, 92.5~
$ DMC   <dbl> 26.2, 35.4, 43.7, 33.3, 51.3, 85.3, 88.9, 145.4, 129.5, 88.0, 88~
$ DC    <dbl> 94.3, 669.1, 686.9, 77.5, 102.2, 488.0, 495.6, 608.2, 692.6, 698~
$ ISI   <dbl> 5.1, 6.7, 6.7, 9.0, 9.6, 14.7, 8.5, 10.7, 7.0, 7.1, 7.1, 22.6, 0~
$ temp  <dbl> 8.2, 18.0, 14.6, 8.3, 11.4, 22.2, 24.1, 8.0, 13.1, 22.8, 17.8, 1~
$ RH    <dbl> 51, 33, 33, 97, 99, 29, 27, 86, 63, 40, 51, 38, 72, 42, 21, 44, ~
$ wind  <dbl> 6.7, 0.9, 1.3, 4.0, 1.8, 5.4, 3.1, 2.2, 5.4, 4.0, 7.2, 4.0, 6.7,~
$ rain  <dbl> 0.0, 0.0, 0.0, 0.2, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,~
$ area  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
```

```
# A tibble: 1 x 8
      n mean_area median_area sd_area min_area max_area zero_count zero_pct
  <int>     <dbl>       <dbl>   <dbl>    <dbl>    <dbl>      <int>    <dbl>
1   517      12.8        0.52    63.7        0    1091.        247     47.8
```

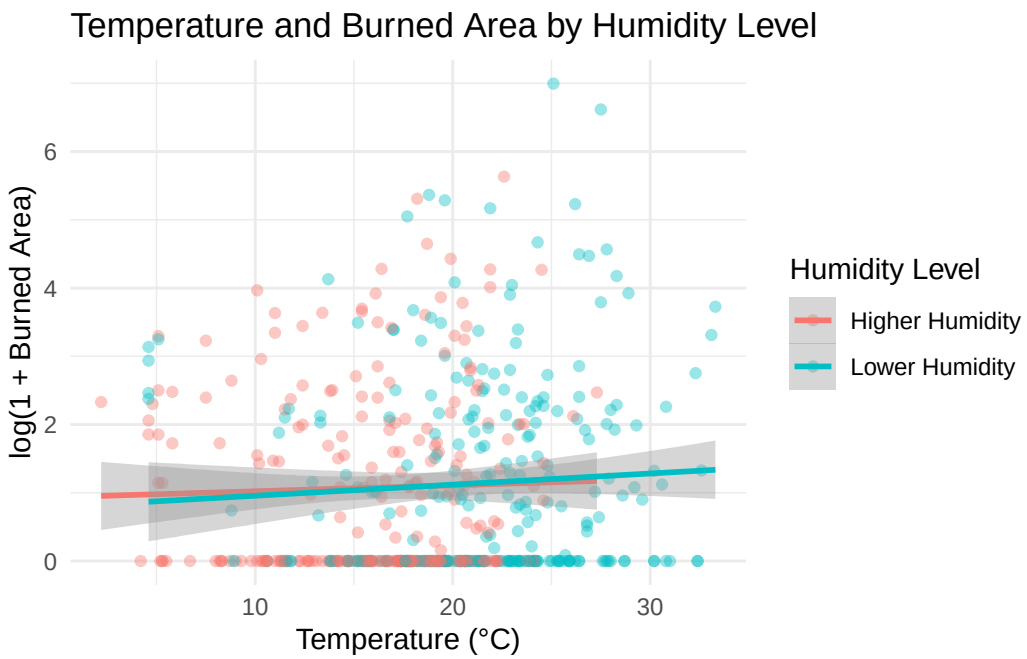
```r
# Create log-transformed response
fires <- fires |>
  mutate(log_area = log1p(area))

# Original burned area
ggplot(fires, aes(x = area)) +
  geom_histogram(bins = 30, fill = "steelblue",
                 color = "white") +
  labs(
    title = "Original Burned Area",
    x = "Burned Area (ha)",
    y = "Count"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 11))
# Log-transformed burned area
ggplot(fires, aes(x = log_area)) +
  geom_histogram(bins = 30, fill = "steelblue",
                 color = "white") +
```

```
  labs(
    title = "Log-Transformed Burned Area",
    x = "log(1 + Area)",
    y = "Count"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 11))
```



The distribution of burned area is highly right-skewed, with most observations concentrated near zero and a small number of extremely large values. After applying a log(area + 1) transformation, the extreme values become less dominant, making the distribution easier to examine. However, a substantial concentration at zero remains, suggesting that the transformed response may still present challenges for linear regression.



The scatter plot suggests a weak positive association between temperature and log-transformed burned area. When observations are divided into higher and lower humidity groups, the fitted regression lines show only small differences in slope, providing limited visual evidence of a temperature–humidity interaction. However, substantial variation in burned area remains at similar temperature levels, and many observations have zero recorded burned area. A formal regression analysis with continuous humidity and additional covariates will be needed to evaluate the interaction effect.

```
# A tibble: 4 x 3
  season     n percent
  <fct>  <int>   <dbl>
1 Winter    31     6
2 Spring    65    12.6
3 Summer   233    45.1
4 Fall     188    36.4
```

Forest fire observations are unevenly distributed across seasons, with most records occurring in summer (45.1%) and fall (36.4%). Winter accounts for only 6.0% of the observations. This seasonal imbalance suggests that fire activity varies considerably across the year and supports including season as an additional covariate in the regression model. However, the smaller number of winter observations may result in less precise estimates for seasonal comparisons.

Team Charter

1. When will you meet as a team to work on the project components? Will these meetings be held in person or virtually?

   We will meet in person twice a week on our statistics class days, specifically Tuesdays and Thursdays at 11:00 AM. If necessary, virtual meetings via Zoom or Google Meet will be arranged to address quick questions or urgent deadlines.

2. What is your group policy on missing team meetings (e.g., how much advance notice should be provided)?

   If a team member cannot attend an in-person meeting, they must give at least 1 hour's advance notice in our group channel. Whenever possible, members unable to attend in person are expected to join virtually.

3. How will your team communicate (email, Slack, text messages)? What is your policy on appropriate response time (within a certain number of hours? Nights/weekends?)?

   Our primary communication channel is Slack. Team members are expected to respond within a few hours during the day, with regular communication taking place until 9:00 PM on weeknights. Urgent matters after 9:00 PM or during weekends can be addressed if agreed upon, but routine responses are generally expected by the following morning.